Solutions to Chapter 11 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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Criteria used. For a family f_a of scalar maps: a *saddle-node* bifurcation occurs at a fixed point $(\bar{a}, \bar{x})$ with $f_x = 1$, $A = \partial f/\partial a \neq 0$ and $D = \partial^2 f/\partial x^2 \neq 0$ (Theorem 11.9); a *period-doubling* bifurcation occurs where the path of fixed points has f_x crossing -1 (Theorem 11.11).

Exercise 11.1

$f_a(x) = a \sin x$. Fixed points: $x = 0$ for all a, plus nonzero solutions of $\sin x/x = 1/a$.

- **Period-doubling.** Along the path $x = 0$, $f_x = a$ crosses -1 at $a = -1$. So there is a period-doubling bifurcation at $(a, x) = (-1, 0)$.
- **Saddle-node.** We need $f_x = a \cos x = 1$ together with $a \sin x = x$, i.e. $\tan x = x$ and $a = 1/\cos x$. The root $x = 0$ gives $a = 1$ but there $A = \sin x = 0$, so that is a pitchfork (the family is odd), not a saddle-node. The first nonzero roots are $x \approx \pm 4.4934$, with $a = 1/\cos x \approx -4.603$. There $A = \sin x \neq 0$ and $D = -a \sin x = -x \neq 0$, so Theorem 11.9 applies: a **saddle-node** at $(a, x) \approx (-4.603, \pm 4.493)$. (Further saddle-nodes occur at the other roots of $\tan x = x$, e.g. $x \approx \pm 7.725$, $a \approx 7.79$.)

Exercise 11.2

$g_a(x) = -x(a + x)$. Fixed points: $x = 0$ and $x = -(a + 1)$; $g_a'(x) = -a - 2x$.

- At $x = 0$, $g_a'(0) = -a$ crosses -1 at $a = 1$: a **period-doubling bifurcation** at $(a, x) = (1, 0)$. (At the other fixed point $g_a' = a + 2 = -1$ for $a = -3$: another one, at $(-3, 2)$.)
- Period-two points: $g_a^2(x) - x = -x\,(x + a + 1)\left(x^2 + (a - 1)x + 1 - a\right)$, so the period-two points solve $x^2 + (a - 1)x + (1 - a) = 0$, with real distinct roots when $a > 1$ (or $a < -3$). For two such points, $x_1 + x_2 = 1 - a$ and $x_1 x_2 = 1 - a$, so the multiplier is

$$g'(x_1)g'(x_2) = a^2 + 2a(x_1 + x_2) + 4x_1x_2 = -a^2 - 2a + 4.$$

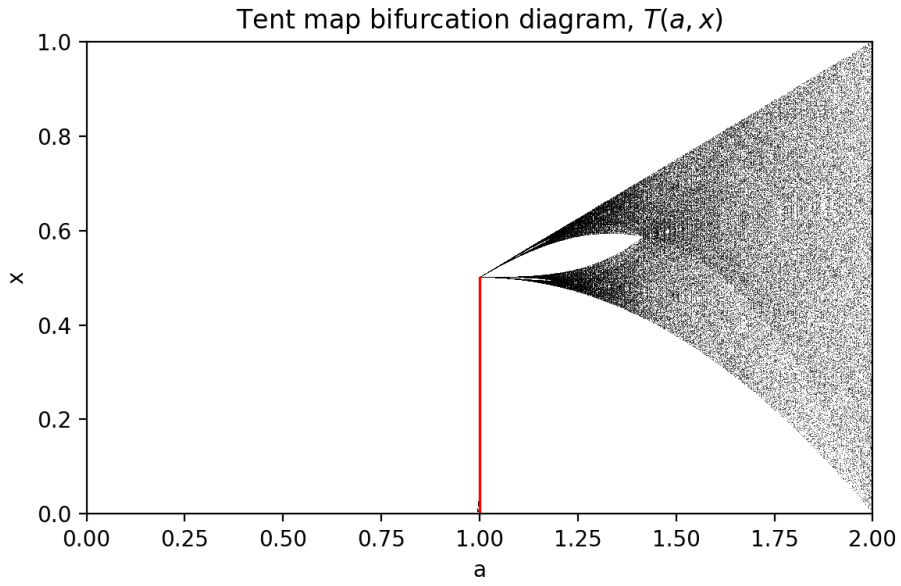
  The period-two orbit is a sink iff $|-a^2 - 2a + 4| < 1$: for $a > 1$ this means $a^2 + 2a - 5 < 0$, i.e. $1 < a < \sqrt{6} - 1$. At $a = \sqrt{6} - 1 \approx 1.449$ the multiplier equals -1, so **the period-two orbit itself period-doubles at** $a = \sqrt{6} - 1$ (to a period-four sink).

Exercise 11.3

$f_a(x) = x^3 - ax$. Fixed points: $x = 0$ and $x = \pm\sqrt{a+1}$ (for $a > -1$); $f_a'(x) = 3x^2 - a$. Along the path $x = 0$, $f_a'(0) = -a$ crosses -1 at $a = 1$. So there is a **period-doubling bifurcation at** $(a, x) = (1, 0)$: for $a > 1$ there is the period-two orbit $\{\pm\sqrt{a-1}\}$ (since $f_a(p) = -p$ gives $p^2 = a - 1$), and f_a is odd so it is symmetric. (At the nonzero fixed points $f' = 2a + 3 = -1$ would need $a = -2$, where they do not exist.) *Remark:* the book's printed answer says $a = -1$, which is the answer for $f_a(x) = x^3 + ax$; for $x^3 - ax$ as printed, the crossing of -1 is at $a = 1$.

Exercise 11.4

$T(a, x) = ax$ for $x \le \frac{1}{2}$, $a(1 - x)$ for $x > \frac{1}{2}$, on $[0, 2] \times [0, 1]$. The diagram below was computed by iterating each a (discarding transients).



Description:

- $0 < a < 1$: $x = 0$ is an attracting fixed point, and every orbit tends to 0.
- $a = 1$: every point of $[0, \frac{1}{2}]$ is fixed (the red segment), a degenerate bifurcation.
- $a > 1$: 0 is repelling, the other fixed point $a/(1 + a)$ has slope $-a$ and is repelling, and every periodic orbit is repelling (multiplier $\pm a^k$). The attractor lies in $[a - \frac{a^2}{2}, \frac{a}{2}]$. For $1 < a < \sqrt{2}$ it consists of 2^m chaotic bands (2 bands for $2^{1/4} < a < \sqrt{2}$, 4 for $2^{1/8} < a < 2^{1/4}$, and so on, accumulating at $a = 1$); the bands merge at $a = 2^{1/2^m}$ (band-merging crises), and for $\sqrt{2} < a \le 2$ it is a single band, filling $[0, 1]$ at $a = 2$. There are no stable periodic orbits, hence no period-doubling cascade or periodic windows.

Exercise 11.5

(a) $f_a(x) = a + \ln x$: $f'(x) = 1/x > 0$, so $f' = -1$ is impossible (no period-doubling). $f' = 1$ at $x = 1$, with $a = x - \ln x = 1$; $A = 1 \neq 0$ and $D = -1/x^2 = -1 \neq 0$. **Saddle-node at** $a = 1$ (at $x = 1$).

(b) $f_a(x) = a - \ln x$: $f' = -1/x < 0$, so no saddle-node. $f' = -1$ at $x = 1$, with $1 = a - 0$: **period-doubling at** $a = 1$.

(c) $f_a(x) = a \ln x$: $f' = a/x$. $f' = 1$ gives $a = x$ and $x = a \ln x$, so $\ln x = 1$, $x = e$, $a = e$; $A = \ln x = 1 \neq 0$, $D = -a/x^2 \neq 0$: **saddle-node at** $a = e$. $f' = -1$ gives $a = -x$ and $\ln x = -1$, $x = 1/e$, $a = -1/e$: **period-doubling at** $a = -1/e$.

(d) $f_a(x) = a + x - e^x$: fixed points $e^x = a$; $f'(x) = 1 - e^x$. $f' = 1$ is impossible ($e^x > 0$), so no saddle-node. $f' = -1$ gives $e^x = 2$, $x = \ln 2$, $a = 2$: **period-doubling at** $a = 2$.

Exercise 11.6

$f_a(x_1, x_2) = (ax_1 - x_2^2,\ x_1 + ax_2)$. Fixed points: $x_1 = (1-a)x_2$ from the second equation; the first then gives $(1-a)x_1 = -x_2^2$, i.e. $(1-a)^2 x_2 = -x_2^2$. So $x_2 = 0$ or $x_2 = -(1-a)^2$, giving two paths: the origin $(0,0)$ and $\big(-(1-a)^3, -(1-a)^2\big)$. They cross at $a = 1$. The Jacobian is $\begin{pmatrix} a & -2x_2 \\ 1 & a \end{pmatrix}$ with characteristic polynomial $(a-\lambda)^2 + 2x_2 = 0$, so $\lambda = 1$ iff $(a-1)^2 + 2x_2 = 0$.

- Along the origin ($x_2 = 0$): $\lambda = a$ (double), so $+1$ is an eigenvalue at $a = 1$.
- Along the other path ($x_2 = -(1-a)^2$): $\lambda = a \pm \sqrt{2}\,|1-a|$, which equals 1 at $a = 1$.

Stability. The origin has both eigenvalues equal to a: it is attracting for $-1 < a < 1$ and repelling for $a > 1$, so it changes from attracting to repelling at $a = 1$. On the other path, near $a = 1$ the eigenvalues are $a \pm \sqrt{2}|1-a|$: one is > 1 and the other is in $(-1, 1)$, so it is a saddle on both sides of $a = 1$; the eigenvalue branch that equals $\sqrt{2} - (\sqrt{2} - 1)a$ passes through $+1$ from above to below, and the other branch passes through $+1$ from below to above. The two paths exchange the unstable direction at $a = 1$ (a transcritical-type crossing).

Exercise 11.7

$f_a(x, y) = (a - x^2 + 0.3y,\ x)$. Fixed points: $y = x$, $x^2 + 0.7x - a = 0$, so $x = \frac{1}{2}\big(-0.7 \pm \sqrt{0.49 + 4a}\big)$. $Df = \begin{pmatrix} -2x & 0.3 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 + 2x\lambda - 0.3$.

- **Saddle-node:** $\lambda = 1$ gives $1 + 2x - 0.3 = 0$, $x = -0.35$, hence $a = x^2 + 0.7x = -0.1225$. Saddle-node at $(a, x, y) = (-0.1225, -0.35, -0.35)$ (where the two fixed points are born, for $a > -0.1225$).
- **Period-doubling:** $\lambda = -1$ gives $1 - 2x - 0.3 = 0$, $x = 0.35$, hence $a = 0.3675$. Period-doubling at $(a, x, y) = (0.3675, 0.35, 0.35)$.

Exercise 11.8

Take $f_a(x) = -(1+a^2)x - x^3$ at $(\bar{a}, \bar{x}) = (0,0)$. Then $x = 0$ is a fixed point for all a and $f_a'(0) = -(1+a^2)$ equals -1 at $a = 0$, but it touches -1 without crossing it ($f_a'(0) \le -1$ on both sides). There is no period-doubling: $f_a = -g_a$ with $g_a(x) = (1+a^2)x + x^3$ odd and increasing with $g_a(x) > x$ for $x > 0$, so $f_a^2(x) = g_a(g_a(x)) > x$ for $x > 0$ (and $< x$ for $x < 0$), and f_a^2 has no fixed points besides 0: there are no period-two orbits near $(0,0)$ for any a.

Exercise 11.9

Let $f_x(\bar{a}, \bar{x}) = -1$. Since $1 - f_x \neq 0$, the implicit function theorem gives a path of fixed points $x(a)$ near $\bar{a}$ with $x(\bar{a}) = \bar{x}$. Differentiating $x(a) = f(a, x(a))$: $x' = f_a + f_x x'$, so $x'(\bar{a}) = \dfrac{f_a}{1 - f_x} = \frac{1}{2} f_a(\bar{a}, \bar{x})$.

(a) The derivative of f_x along the path is

$$\frac{d}{da} f_x\big(a, x(a)\big) = f_{xa} + f_{xx}\, x'(a),$$

and at $a = \bar{a}$ this equals $f_{ax} + \frac{1}{2} f_{xx} f_a$, the left side of (11.22) (read as $\neq 0$). So if (11.22) holds, the total mixed derivative $\frac{d^2 f}{da\, dx}(a, x(a))$ is nonzero at $a = \bar{a}$, and by continuity for all a near $\bar{a}$.

(b) Any fixed point of f near $(\bar{a}, \bar{x})$ lies on the path $(a, x(a))$ (the path is the full set of fixed points near $(\bar{a}, \bar{x})$ since $f_x \neq 1$). The function $\varphi(a) = f_x(a, x(a))$ has $\varphi(\bar{a}) = -1$ and $\varphi'(\bar{a}) \neq 0$, so φ is strictly monotone near $\bar{a}$ and $\varphi(a) \neq -1$ for $a \neq \bar{a}$. Hence there is a neighborhood of $(\bar{a}, \bar{x})$ with no other fixed point satisfying $f_x = -1$: period-doubling bifurcations are isolated.

Exercise 11.10

(a) $f(x) = x + cx^3 = x(1 + cx^2)$. If $c < 0$, then for small $x \neq 0$ we have $0 < 1 + cx^2 < 1$, so f preserves the sign of x and $|f(x)| < |x|$: orbits decrease monotonically in magnitude to 0. Thus 0 is attracting. If $c > 0$, then $|f(x)| > |x|$ for all $x \neq 0$ and orbits move away: 0 is repelling.

(b) Let $g(x) = -x + bx^2 + dx^3 + O(x^4)$. Then 0 is attracting for g iff it is attracting for g^2 (if $g^{2n}(x) \to 0$ then $g^{2n+1}(x) = g(g^{2n}(x)) \to g(0) = 0$). Compute $g(g(x))$ with $y = g(x) = -x + bx^2 + dx^3$: $y^2 = x^2 - 2bx^3 + O(x^4)$, $y^3 = -x^3 + O(x^4)$, so

$$g(y) = -y + by^2 + dy^3 = x - bx^2 - dx^3 + bx^2 - 2b^2x^3 - dx^3 + O(x^4) = x - 2(b^2 + d)x^3 + O(x^4).$$

By (a) (the $O(x^4)$ terms do not affect the sign), 0 is attracting for g^2, hence for g, iff $-2(b^2 + d) < 0$, i.e. $b^2 + d > 0$; and repelling iff $b^2 + d < 0$.

Exercise 11.11

$f_a(x) = (x-a)(2x-3a) + x + c$. Fixed points: $2x^2 - 5ax + 3a^2 + c = 0$, i.e. $x_\pm = \dfrac{5a \pm s}{4}$ with $s = \sqrt{a^2 - 8c}$ (they exist iff $a^2 \geq 8c$). Also $f_a'(x) = 4x - 5a + 1 = 1 \pm s$ at $x_\pm$.

- $c > 0$**:** fixed points exist only for $|a| \geq 2\sqrt{2c}$, so there are **saddle-node** bifurcations (where $s = 0$, $f' = 1$) at $a = \pm 2\sqrt{2c}$, $x = 5a/4 = \pm\frac{5}{2}\sqrt{2c}$ (here $A = \partial f/\partial a$ is nonzero). For larger $|a|$, x_+ is repelling ($f' = 1 + s > 1$) and x_- is attracting while $0 < s < 2$, i.e. $8c < a^2 < 8c + 4$. At $s = 2$, i.e. $a = \pm\sqrt{8c+4}$, $f'(x_-) = -1$: **period-doubling** at $(a, x) = \big(\pm\sqrt{8c+4},\, (5a-2)/4\big)$.
- $-\frac{1}{2} < c < 0$**:** $a^2 - 8c > 0$ always, so there is no saddle-node; x_+ is always repelling, x_- is attracting for $|a| < \sqrt{8c+4}$ (as $s < 2$), and there are **period-doublings** at $a = \pm\sqrt{8c+4}$, $x = (5a-2)/4$.
- $c = -\frac{1}{2}$**:** $f'(x_-) = 1 - \sqrt{a^2+4} \leq -1$ with equality only at $a = 0$: it touches -1 but does not cross, so there is no bifurcation. The same holds for $c < -\frac{1}{2}$.
- $c = 0$ **(special):** $s = |a|$, the fixed points are $x = a$ (with $f' = 1 - a$) and $x = \frac{3}{2}a$ (with $f' = 1 + a$): two lines of fixed points crossing at $(0,0)$, a **transcritical** bifurcation (exchange of stability, $f' = 1$ at $a = 0$) at $(0,0)$, and period-doublings at $a = 2$ ($x = 2$) and $a = -2$ ($x = -3$).

So there is a bifurcation exactly when $c > -\frac{1}{2}$.

Exercise 11.12

For $\dot v = f_a(v)$ with a supercritical Hopf bifurcation (Figure 11.23): for $a < 0$ the equilibrium 0 is a sink; at $a = 0$ it loses stability; for $a > 0$ it is a source surrounded by a *stable* limit cycle of radius $\propto \sqrt{a}$. For $\dot v = -f_a(v)$ time is reversed, which interchanges stable and unstable: for $a < 0$ the equilibrium is a *source*; for $a > 0$ it is a *sink* surrounded by an *unstable* limit cycle of radius $\propto \sqrt{a}$, which shrinks onto the equilibrium as $a \to 0^+$. In the bifurcation diagram (amplitude versus a) the branch $r = \sqrt{a}$ is the same curve but now drawn dashed (unstable), with the equilibrium line solid for $a > 0$ and dashed for $a < 0$; in the phase portraits the arrows on all orbits are reversed. This is a **subcritical** Hopf bifurcation (the bifurcating orbits are unstable and exist on the side where the equilibrium is stable).

Exercise 11.13

The Chua equations are $\dot x = c_1(y - x - g(x))$, $\dot y = c_2(x - y + z)$, $\dot z = -c_3 y$, with $c_1 = 15.6$, $c_2 = 1$, $m_0 = -\frac{8}{7}$, $m_1 = -\frac{5}{7}$. The outer equilibria are $\pm\big(\frac{m_1 - m_0}{m_1+1}, 0, -\frac{m_1-m_0}{m_1+1}\big) = \pm(\frac{3}{2}, 0, -\frac{3}{2})$, where $|x| > 1$ and $g'(x) = m_1$. Let $\alpha = c_1(1 + m_1) = 15.6 \cdot \frac{2}{7} = \frac{156}{35}$. The Jacobian there is

$$\begin{pmatrix} -\alpha & c_1 & 0 \\ 1 & -1 & 1 \\ 0 & -c_3 & 0 \end{pmatrix}, \qquad \lambda^3 + (1+\alpha)\lambda^2 + (c_3 + \alpha - c_1)\lambda + \alpha c_3 = 0.$$

By Exercise T11.21(a), the roots are $-a_2$ and $\pm i\sqrt{a_1}$ iff $a_1a_2 = a_0$ and $a_1 > 0$. Here $a_2 = 1+\alpha$, $a_1 = c_3 + \alpha - c_1$, $a_0 = \alpha c_3$, so $a_1a_2 - a_0 = c_3 + (\alpha - c_1)(1+\alpha) = 0$, i.e.

$$c_3 = (c_1 - \alpha)(1+\alpha) = \frac{78}{7}\cdot\frac{191}{35} = \frac{14898}{245} \approx \mathbf{60.81}.$$

At this value $a_1 \approx 49.7 > 0$, the eigenvalues are $\pm i\sqrt{a_1} \approx \pm 7.05\,i$ and $-a_2 \approx -5.46$, and $a_1a_2 - a_0 = c_3 - 60.81$ changes sign linearly, so the pair crosses the imaginary axis with nonzero speed (Routh–Hurwitz: stable for $c_3 > 60.81$, unstable below). So a Hopf bifurcation occurs at $c_3 \approx 60.81$, consistent with Figure 9.10(a) ($c_3 = 50$) showing two periodic orbits.

Exercise 11.14

The system is $\dot{x} = r\big(\frac{c}{x} - d\big) + ax - bxy$, $\dot{y} = r\big(b - \frac{a}{y}\big) - cy + dxy$.

(a) At $(x, y) = (c/d, a/b)$: $\frac{c}{x} - d = 0$ and $a - by = 0$, so $\dot{x} = 0$; $b - \frac{a}{y} = 0$ and $dx - c = 0$, so $\dot{y} = 0$. It is an equilibrium for all r.

(b) The Jacobian at the equilibrium is

$$J = \begin{pmatrix} -r\,d^2/c & -bc/d \\ da/b & r\,b^2/a \end{pmatrix}, \quad \operatorname{tr} J = r\Big(\frac{b^2}{a} - \frac{d^2}{c}\Big), \quad \det J = ac - r^2\frac{b^2d^2}{ac}.$$

At $r = 0$ the eigenvalues are $\pm i\sqrt{ac}$ (pure imaginary, with nonzero imaginary part). Near $r = 0$ the eigenvalues are $\frac{1}{2}\operatorname{tr} J \pm i\sqrt{\det J - \frac{1}{4}(\operatorname{tr} J)^2}$, with real part $\frac{r}{2}\big(\frac{b^2}{a} - \frac{d^2}{c}\big)$ crossing 0 at $r = 0$ with nonzero speed $\frac{1}{2}\big(\frac{b^2}{a} - \frac{d^2}{c}\big)$ provided $b^2c \neq ad^2$ (assumed here). The equilibrium has dimension 2, so there are no other eigenvalues, and the hypotheses of Theorem 11.21 hold: a Hopf bifurcation at $r = 0$. (At $r = 0$ every orbit near the equilibrium is periodic, since the predator–prey system has the conserved quantity $H = -c\ln x + dx - a\ln y + by$, so, as in Example 11.24, the whole family of periodic orbits exists only at $r = 0$.)

(c) A path of periodic orbits that converges to the equilibrium can exist only at a parameter value where the equilibrium is not hyperbolic, i.e. where Df has an eigenvalue pair on the imaginary axis (otherwise the equilibrium is a sink, source or saddle, and no periodic orbits lie in a neighborhood of it). That requires $\operatorname{tr} J = r\big(\frac{b^2}{a} - \frac{d^2}{c}\big) = 0$, i.e. $r = 0$ (for $b^2c \neq ad^2$). So all periodic orbits bifurcating from the equilibrium occur at $r = 0$. (As a check, along solutions $\dfrac{dH}{dt} = r\Big[\big(b - \frac{a}{y}\big)^2 - \big(d - \frac{c}{x}\big)^2\Big]$, which vanishes identically for $r = 0$.) If $b^2c = ad^2$ the trace vanishes for all r and the hypotheses of the theorem fail.